



# CV Analyzer - Complete Setup Guide



## Project Structure



```
cv-analyzer-project/
├── cv_analyzer_server.py      # Main MCP server (THE ONLY PYTHON FILE YOU NEED)
├── requirements.txt          # Python dependencies
├── SETUP_GUIDE.md            # This file
├── USAGE_EXAMPLES.md         # Usage examples
├── test_cvs/                 # Folder where you put CVs to analyze
│   ├── candidate1.pdf
│   ├── candidate2.docx
│   ├── candidate3.txt
│   └── ...
└── outputs/                  # Folder for exported reports
    └── analysis_report.json
```

**IMPORTANT:** You only need **ONE Python file** (`cv_analyzer_server.py`) - everything is included in it!



## Step-by-Step Setup

### Step 1: Install Ollama (Free Local LLM)

#### Windows:

1. Download from: <https://ollama.com/download>
2. Run installer
3. Ollama will start automatically

#### Verify installation:



bash

```
ollama --version
```

#### Pull the recommended model:



bash

```
# Best for CV analysis (recommended)
ollama pull qwen2.5:14b
```

```
# Alternative if you have less RAM
ollama pull llama3.1:8b
```

Test the model:



bash

```
ollama run qwen2.5:14b "Hello, are you working?"
```

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Step 2: Create Project Folder



bash

```
# Create main folder
mkdir cv-analyzer-project
cd cv-analyzer-project
```

```
# Create subfolders
mkdir test_cvs
mkdir outputs
```

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Step 3: Install Python Dependencies

Option A: Using pip directly



bash

```
pip install mcp PyPDF2 python-docx ollama-python
```

Option B: Using requirements.txt (recommended)



bash

```
pip install -r requirements.txt
```

Verify installation:



bash

```
python -c "import mcp, PyPDF2, docx, ollama; print('✅ All packages installed!')"
```

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Step 4: Configure MCP in Claude Desktop

Find your Python path:



bash

```
# Windows
where python

# Output example:
# C:\Python311\python.exe
```

Update Claude Desktop MCP config:

Location: %APPDATA%\Claude\claude\_desktop\_config.json



json

```
{
  "mcpServers": {
    "cv-analyzer": {
      "command": "C:\\Python311\\python.exe",
      "args": [
        "C:\\Users\\YourName\\cv-analyzer-project\\cv_analyzer_server.py"
      ]
    }
  }
}
```

**Replace:**

- C:\\Python311\\python.exe → Your Python path
- C:\\Users\\YourName\\cv-analyzer-project\\ → Your project folder path

**IMPORTANT:** Use double backslashes \\ in Windows paths!

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**Step 5: Test the Server**

Before connecting to Claude, test the server directly:



```
bash

cd cv-analyzer-project
python cv_analyzer_server.py
```

You should see:



 CV Analyzer MCP Server Starting...  
 Waiting for connections...

Press Ctrl+C to stop.

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**Step 6: Restart Claude Desktop**

1. Completely close Claude Desktop
  2. Reopen it
  3. Look for the  icon (MCP tools available)
-



# Quick Usage Test

## Test 1: Analyze a Single CV

- 1. Put a CV in test\_cvs/ folder (e.g., john\_doe.pdf)
- 2. In Claude, say:



Analyze this CV:

Path: C:\Users\YourName\cv-analyzer-project\test\_cvs\john\_doe.pdf

Job Description:

"We need a Senior Python Developer with 5+ years experience.  
Required: Python, FastAPI, PostgreSQL, Docker.  
Preferred: AWS, Kubernetes."

## Test 2: Batch Analyze All CVs



Analyze all CVs in:

C:\Users\YourName\cv-analyzer-project\test\_cvs

Job Description:

"Full-stack developer. React, Node.js, MongoDB required."

## Test 3: Get Rankings



Show me the top 5 candidates ranked by score



# Troubleshooting

## ✗ "Ollama not found"

Check if Ollama is running:



bash

ollama list

**If not running, start it:**



bash

*# Windows: Ollama should auto-start*

*# Or run: ollama serve*

**Verify model is downloaded:**



bash

ollama list

*# Should show: qwen2.5:14b or llama3.1:8b*

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**✗ "Module not found: mcp"**



bash

pip **install** --upgrade mcp

---

**✗ "Module not found: PyPDF2"**



bash

pip **install** PyPDF2 python-docx ollama-python

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**✗ "File not found" errors**

**Use absolute paths:**



✓ Good: C:\Users\Name\cv-analyzer-project\test\_cvs\cv.pdf

✗ Bad: test\_cvs/cv.pdf

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## ✗ "JSON parse error from LLM"

**Solution 1:** The model might be struggling. Try a different one:



bash

```
ollama pull mistral
```

Then edit line 122 in cv\_analyzer\_server.py:



python

```
def __init__(self, llm_model: str = "mistral"):
```

**Solution 2:** Check Ollama is running:



bash

```
ollama ps  
# Should show active models
```

---

## ✗ MCP Server not connecting to Claude

1. Check config file syntax (valid JSON, double backslashes)
2. Verify Python path:



bash

```
where python
```

3. Check server file path exists
  4. Restart Claude Desktop completely
  5. Check Claude logs:
    - Windows: %APPDATA%\Claude\logs\
-

# System Requirements

## Minimum:

- **RAM:** 8GB (use llama3.1:8b)
- **Storage:** 10GB free (for models)
- **OS:** Windows 10/11, macOS, Linux
- **Python:** 3.8+

## Recommended:

- **RAM:** 16GB+ (use qwen2.5:14b)
- **CPU:** Multi-core processor
- **Storage:** SSD for faster loading

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## Model Comparison

| Model       | Size  | RAM  | Speed  | Quality |
|-------------|-------|------|--------|---------|
| qwen2.5:14b | 8.5GB | 16GB | Medium | ★★★★★   |
| llama3.1:8b | 4.7GB | 8GB  | Fast   | ★★★★    |
| mistral:7b  | 4.1GB | 8GB  | Fast   | ★★★★    |
| phi3:medium | 7.9GB | 16GB | Fast   | ★★★     |

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## Privacy & GDPR

### What This Tool Does:

- ✓ **Local processing** - CVs never leave your machine
- ✓ **Automatic anonymization** - Emails, phones, addresses removed
- ✓ **No cloud APIs** - Everything runs on your computer
- ✓ **No data training** - Models don't learn from your data

### What Gets Anonymized:

- ✓ Email addresses → [EMAIL]
- ✓ Phone numbers → [PHONE]
- ✓ URLs → [URL]
- ✓ Street addresses → [ADDRESS]
- ✓ Postal codes → [POSTAL]
- ✓ Social media handles → [SOCIAL]
- ✓ LinkedIn/GitHub profiles → [LINKEDIN]/[GITHUB]

### Candidate Consent Template:

"Your CV will be analyzed using AI technology to assess job fit. Personal information (email, phone, address) will be automatically removed before analysis. All processing happens locally on our systems—your data never leaves our infrastructure and is not used to train any AI models. Results are stored temporarily for recruitment purposes only. You can request deletion of your data at any time."

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# Getting Help

## Common Issues Checklist:

- ☐ Ollama is installed and running
- ☐ Correct model is downloaded (`ollama list`)
- ☐ Python packages installed (`pip list | grep mcp`)
- ☐ Config file has correct paths (double backslashes!)
- ☐ Claude Desktop restarted after config change
- ☐ Test CVs are in supported formats (PDF, DOCX, TXT)

## Still Having Issues?

1. **Test server directly:** `python cv_analyzer_server.py`
  2. **Check Claude logs:** `%APPDATA%\Claude\logs\`
  3. **Verify Ollama:** `ollama ps`
  4. **Test model:** `ollama run qwen2.5:14b "test"`
- 

## Next Steps

Once everything is working:

1. ☒ Put your CVs in `test_cvs/` folder
2. ☒ Prepare your job description
3. ☒ Run batch analysis
4. ☒ Review rankings
5. ☒ Export report

See `USAGE_EXAMPLES.md` for detailed examples!

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## Tips for Best Results

### Job Descriptions:

- Be specific about required vs. preferred skills
- Include years of experience needed
- Mention key technologies/tools
- List important soft skills

### CV Files:

- Use clear filenames (e.g., `john_smith_senior_dev.pdf`)
- Ensure text is extractable (not scanned images)
- Supported formats: PDF, DOCX, TXT
- One CV per file

### Analysis:

- Run batch analysis for multiple candidates
- Use rankings to shortlist top candidates
- Use `compare_candidates` for final selection
- Export reports for record-keeping

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 **Setup Complete! You're ready to analyze CVs.**